

YUSUF ZAIDANY ARRAIHAN

Supply Chain Analytics | Demand & Inventory Planning | Process Mining, Data, Agentic AI & Optimisation

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SUMMARY

Supply chain analytics professional with a Bachelor of Management in Operations and Supply Chain and two years of experience turning operational data into decisions. Works across the full analytics loop: mapping how a process actually runs, mining the event data behind it, forecasting demand, optimising inventory and network decisions, and automating the reporting that used to be manual. Builds this in Python and SQL with process mining (Celonis, pm4py), optimisation, and simulation, and presents it through Power BI and Streamlit. Domain grounding in procurement, medical device distribution, and regulated healthcare operations. Certified in Process Mining by Eindhoven University of Technology, with completed specialisations in Supply Chain Analytics, Supply Chain Management from Rutgers Business School, and IT Automation with Python. English at IELTS C2, German B1.

EXPERIENCE

Supply Chain & Operations Analytics, Contra.com

Sep 2024 - Present

Remote, international clients

- Analysed operational event data and mapped end-to-end processes for international clients, cutting cycle times by an average of 22% and removing around 14 hours of manual work per week.
- Applied process mining (Celonis) to real client event logs to quantify bottlenecks, rework loops and process variants that manual review had missed.
- Built automated data and reporting pipelines in Python to replace manual data entry and recurring reporting.
- Translated analytical findings into operational decisions with client stakeholders and managed delivery end to end.

Operations Manager, HIDATA Digital Technology, Project-based

Feb 2025 - Aug 2026

Surabaya, Indonesia (Hybrid)

- Audited internal datasets using SQL to resolve data quality issues and prepare model-ready inputs for demand and performance prediction, achieving 98% data accuracy.
- Translated business operational requirements into technical specifications for a machine learning implementation.
- Mapped internal workflows with cross-functional technical teams and identified bottlenecks that cut operational waste by 19%.

Operations & Systems Analyst, Tuju Satu Studio

Jan 2024 - Feb 2025

Surabaya, Indonesia

- Coordinated procurement and supply chain logistics for three government infrastructure projects, aligning technical requirements and delivery schedules with municipal and health-institutional stakeholders.
- Designed operational workflow and resource allocation strategy for high-stakes facilities, including a high-containment biosafety laboratory.
- Mapped and rebuilt procedural workflows to remove operational bottlenecks and meet national regulatory requirements.

Operations & Procurement Intern, PT. Artha Genetik Lab Indonesia

Feb 2024 - Jul 2024

Yogyakarta, Indonesia (medical device importer and distributor)

- Coordinated end-to-end procurement and supply chain tracking for specialised medical equipment distributed to healthcare facilities, working with international vendors under import and distribution controls.
- Monitored vendor lead times and distribution timelines using data tools.
- Resolved logistics bottlenecks to maintain inventory continuity for regulated medical goods.

Research Assistant, Universitas Airlangga

May 2023 - Dec 2023

Surabaya, Indonesia

- Conducted data modelling and operational research on supply chain resilience and service quality improvement.
- Scraped, cleaned and synthesised operational datasets from public health clinics into analytical reports that formed the groundwork for an academic journal submission.

Quality Assurance Intern, Ministry of Health of the Republic of Indonesia

Nov 2022 - Apr 2023

Central Java, Indonesia

- Audited operational processes and digital health data frameworks against national quality standards.
- Reviewed patient records and medical documentation to monitor institutional healthcare workflows and system quality metrics.

SUPPLY CHAIN ANALYTICS PROJECTS

★ Supply Chain Analytics Compendium, Six Analytical Cases

2026

Regression, forecasting, ABC classification, push-pull economics, sourcing risk for semiconductor industry

- Worked six supply chain problems end to end and documented each as a case: trend and multiple regression forecasting on a 24-month series (R-squared 0.439 to 0.948, out-of-sample MAPE 10.57%), ABC classification across 36 SKUs, push versus pull inventory economics, and sourcing analysis under supplier concentration.
- Reported weak fits and contradicted rules of thumb rather than only favourable results, including a statistically significant trend model rejected as unusable for planning because its standard error reached half the mean order size.

Google Drive:  [Supply_Chain_Analytics_Compendium.pptx](#)

★ Inventory Brain & Demand Planning Platform

2026

Python, forecasting, optimisation, discrete-event simulation, MLOps, dashboard

- End-to-end planning platform combining demand forecasting, multi-echelon inventory optimisation across distribution tiers, discrete-event simulation of replenishment policies under demand and lead-time variability, and a decision dashboard for scenario comparison.

GitHub: github.com/yzarrai/inventory-brain

★ Demand & Stock Fluctuation Forecasting Machine Learning

2026

Python, SARIMA, Prophet, XGBoost, SHAP

- Benchmarked SARIMA and Prophet against XGBoost under walk-forward validation on MAE, RMSE and MAPE, then converted the selected forecast into service-level-driven safety stock and reorder point recommendations, with SHAP attribution to explain which demand drivers moved each prediction.

GitHub: github.com/yzarrai/retail-demand-forecasting-ml

Procurement RAG Assistant

2026

Python, hybrid search (BM25 + Dense RRF), LLM grounding

- Built an enterprise-grade retrieval-augmented generation assistant for procurement documents with clause-level citations, a 5-layer grounding guardrail system to prevent hallucination, and structured commercial risk scanning across contract text.

GitHub: github.com/yzarrai/procure-rag

Stochastic Cold-Chain Distribution Planner

2026

Python, MILP, stochastic optimization

- Developed a two-stage stochastic optimization and decision-support pipeline for cold-chain vaccine distribution, modeling maritime disruption scenarios to recommend robust distribution plans under uncertainty.

GitHub: github.com/yzarrai/stochastic-coldchain-planner

Multi-Agent Supply Chain Agentic Artificial Intelligence

2025

Python, MILP (PuLP), SARIMAX, DuckDB, ChromaDB, FastAPI, Docker

- Built a 4-agent autonomous swarm (Demand, Inventory, Risk, Logistics) using a 2-round multi-agent debate protocol to reach Pareto-optimal trade-offs, combined with MILP stochastic lot-sizing, SARIMAX forecasting, and a Monte Carlo digital twin for disruption stress-testing.
- Integrated a DuckDB/ChromaDB hybrid RAG data layer with a generative control tower dashboard supporting human-in-the-loop approvals and simulated SAP/Oracle ERP webhook dispatch, validated by 90 automated tests.

GitHub: github.com/yzarrai/supply-chain-agentic-ai

Supply Chain Simulation LLM Agent

2025

Python, SimPy, LangChain, Pydantic

- Built a discrete-event simulation (SimPy) of a 4-echelon supply chain with (s, Q) replenishment policy, using Common Random Numbers to cut Monte Carlo variance by 19–37% for statistically policy comparisons.
- Designed a zero-hallucination LangChain agent constrained to strict tool schemas so every quantitative claim is grounded in simulation output, validated with 39+ automated tests.

GitHub: github.com/yzarrai/supply-chain-simulation-agent

Decision Support & MLOps Systems (Banking, Clinic Operations)

2025

Python, SimPy, pm4py, MLflow, discrete-event simulation

- Built three end-to-end decision support and MLOps systems across loan origination and clinic operations, combining discrete-event simulation, process mining, ML forecasting, and production MLOps (MLflow registry gates, drift monitoring, closed-loop retraining).

GitHub: github.com/yzarrai/loanflow-dss/
github.com/yzarrai/loanflow-mlops/ / github.com/yzarrai/clinic-flow-dss/

EDUCATION

Universitas Airlangga

2018 - 2023

Bachelor of Management, Operations and Supply Chain / Operations Research

Grade: 3.60/4.00 High Distinction (Cum Laude)

Thesis: Analysis of Service Quality Improvement Using the DMAIC Method at Hortus Medicus Clinic

- Measured a total patient lead time of 534 minutes with 80% classified as non-value-added, and identified a 182-minute post-lab re-queue caused by manual transfer of lab results as the primary bottleneck.
- Designed a future-state process projected to remove 42.6% of non-value-added time and raise process cycle efficiency from 20.0% to 30.3%.

CERTIFICATIONS

- Supply Chain Analytics Specialization (6 courses): Analytics Essentials, Business Intelligence & Competitive Analysis, Demand Analytics, Inventory Analytics, Supply Chain Analytics, Sourcing Analytics
- Supply Chain Management Specialization, Rutgers Business School (5 courses): Logistics, Operations, Planning, Sourcing, Management Strategy. Capstone: Medical Technology Corporation
- Process Mining: Data Science in Action, Eindhoven University of Technology
- Health Informatics Specialization, Johns Hopkins University
- Google IT Automation with Python Professional Certificate, Google
- AI and cloud: Google Artificial Intelligence Professional Certificate; LangChain for LLM Application Development and Agentic AI, DeepLearning.AI; Fundamentals of Cloud & Generative AI, Amazon Web Services
- Full Stack Product Management, RevouU

CURRENTLY STUDYING

- APICS CSCP (Certified Supply Chain Professional), target completion: 2027
- SAP S/4HANA and HL7 FHIR

SKILLS

Supply Chain Analytics: Demand planning and forecasting, inventory optimisation, safety stock and reorder point modelling, sourcing and supplier risk analytics, network and facility location optimisation, supply chain KPIs

Supply Chain Operations: Procurement, inventory management, logistics coordination, regulated goods handling, Lean Six Sigma (DMAIC), business process management, BPMN

Data & Analytics: Python, SQL, Power BI, Tableau, Streamlit, statistical modelling, time series forecasting, machine learning, discrete-event simulation, process mining (Celonis, pm4py)

Optimisation & Automation: Linear and integer programming, p-median facility location, workflow automation, Python automation, MLOps (MLflow, FastAPI, drift monitoring), CI/CD with GitHub Actions, n8n, Zapier, Make.com

Enterprise Systems: Working knowledge of SAP, Oracle, Odoo and Salesforce (CRM) from a solutions and implementation perspective

Project Management: Jira, Notion

CRM: Salesforce, HubSpot

Languages: Indonesian (native), English (IELTS C2), German (Duolingo B1)

HONORS & AWARDS

- University Ambassador 2021 (Duta Universitas Airlangga)
- KBMI (P2MW) Funding Awardee